



the full report

-25%

deliverability at intervals multiple of 300 ms

0

payload bytes ever corrupted

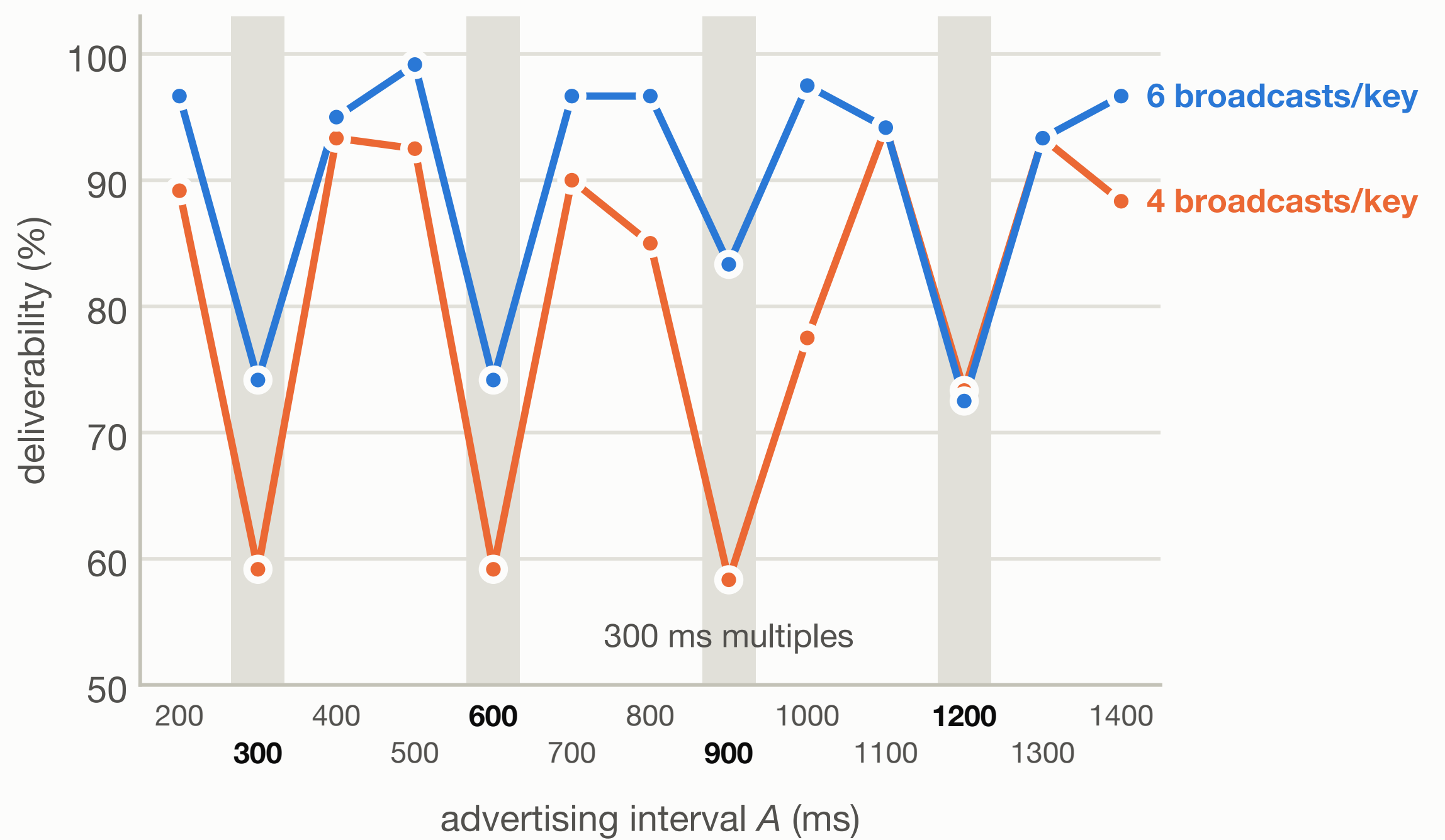
158s

median broadcast-to-report delay

Deliverability collapses at every multiple of 300 ms

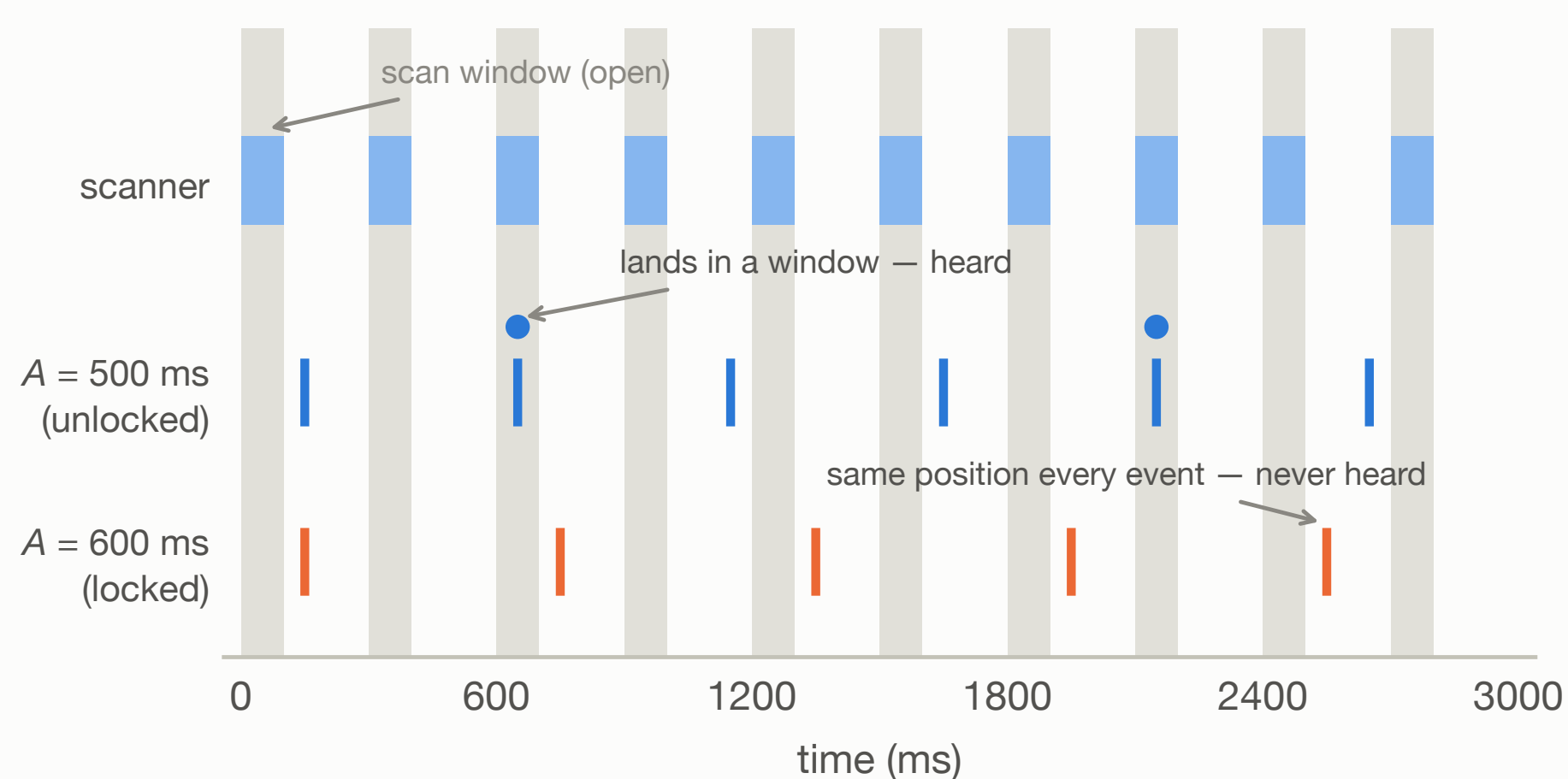
- Sharp dips at 300, 600, 900, 1200 ms
- **69.3%** there against **92.7%** elsewhere
- Beyond chance: $p < 5 \times 10^{-5}$
- Same dips at 4 and 6 broadcasts per key
- So it tracks the interval, not key rotation

More broadcasts cannot rescue it — at 16 per key, 98.9% at 500 ms against 76.7% at 600 ms

Advertising interval A , 200–1400 ms at -24 dBm.

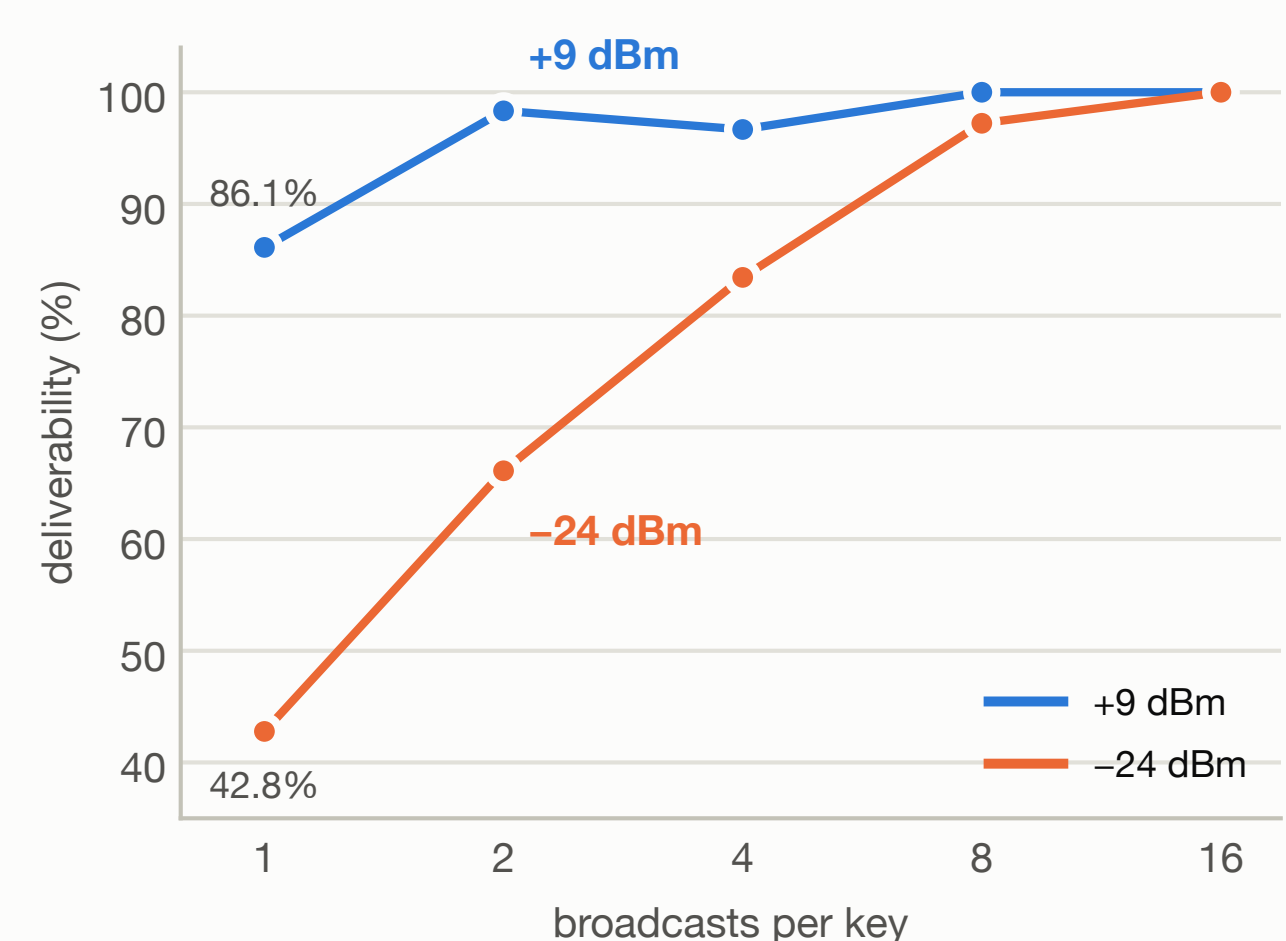
Why: scan-window arithmetic

- A scanner listens once per ~ 300 ms cycle, deaf between
- A multiple of 300 ms lands in the same slot every time
- Stuck in the deaf gap, every broadcast misses alike

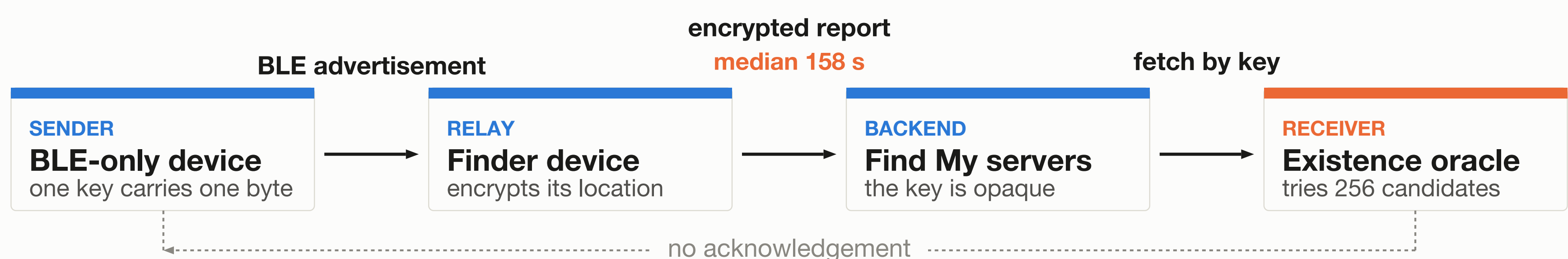


Repetition versus power

- Broadcasts per key: the strongest control
- 42.8% at one broadcast, 100% at sixteen
- Power moves the ceiling, not the shape

Broadcasts per key, at -24 dBm and $+9$ dBm.

The channel, end to end



- One byte per key: HKDF under a 32-byte uid , carrier $x(d \cdot G)$
- Receiver tries 256 candidates; the one with a report wins

- Every carrier is valid — a miss is loss, never corruption
- The uid keeps the channel private end to end

What the channel is

- Erasure channel — never corrupts
- At least 8 broadcasts per key
- The nulls expose the scanners' schedule
- 99.5–99.94% delivered at full power
- Never near a multiple of 300 ms
- Delay-tolerant only — never real-time